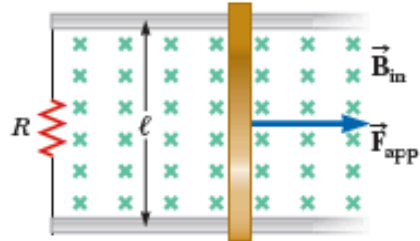


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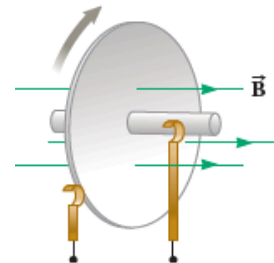
Page 818, Task 13.

A metal rod of mass m slides without friction along two parallel horizontal rails, separated by a distance l , and connected by a resistor R , as shown in the figure. A uniform vertical magnetic field of magnitude B is applied perpendicular to the plane of the paper. The applied force shown in the figure acts only for a moment, to give the rod a speed v . In terms of m , l , R , B , and v , find the distance the rod will then slide as it coasts to a stop.



Page 818, Task 17.

A homopolar generator will have a magnetic field of magnitude $B = 0.900$ T and the radius of the disk is $r = 0.400$ m. The desired emf to be generated with the device is $\varepsilon = 25.0$ V. Your supervisor asks you to determine the required angular speed of the disk to achieve this result.



Page 820, Task 25.

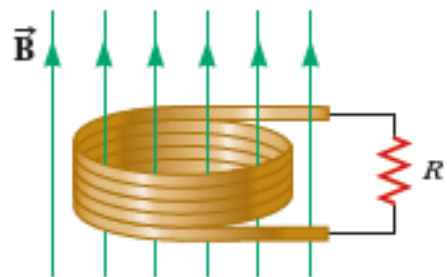
The rotating loop in an AC generator is a square 10.0 cm on each side. It is rotated at 60.0 Hz in a uniform magnetic field of 0.800 T.

Calculate

- the flux through the loop as a function of time,
- the emf induced in the loop,
- the current induced in the loop for a loop resistance of $1.00\ \Omega$,
- the power delivered to the loop,
- the torque that must be exerted to rotate the loop.

Page 820, Task 31.

A circular coil enclosing an area of 100 cm^2 is made of 200 turns of copper wire (the figure). The wire making up the coil has no resistance; the ends of the wire are connected across a $5.00\text{-}\Omega$ resistor to form a closed circuit. Initially, a 1.10-T uniform magnetic field points perpendicularly upward through the plane of the coil. The direction of the field then reverses so that the final magnetic field has a magnitude of 1.10 T and points



downward through the coil. If the time interval required for the field to reverse directions is 0.100 s, what is the average current in the coil during that interval?

Page 822, Task 42.

Review. In the figure, a uniform magnetic field decreases at a constant rate $dB/dt = -K$, where K is a positive constant. A circular loop of wire of radius a containing a resistance R and a capacitance C is placed with its plane normal to the field.

- (a) Find the charge Q on the capacitor when it is fully charged.
- (b) Which plate, upper or lower, is at the higher potential?
- (c) Discuss the force that causes the separation of charges.

